DL Practical 1

Boston Housing Price Prediction using Linear Regression & Deep Neural Network

Aim

Predict Boston house prices using Linear Regression with Deep Learning concepts.

Concept Explanation

This practical is based on **supervised learning**.

The model learns the relationship between:

- Input features (crime rate, rooms, tax, etc.)
- Output value (house price)

The dataset contains:

- 506 rows
- 13 input features
- 1 target column (Price)

Important Features in Dataset

Feature	Meaning
CRIM	Crime rate
RM	Number of rooms
AGE	Old houses proportion
TAX	Property tax
PTRATIO	Student-teacher ratio
LSTAT	Lower income population percentage

More rooms generally increase price, while crime rate lowers price.

What is Linear Regression?

Linear Regression predicts output using a straight-line equation.

The model tries to find:

y=θ1+θ2xy = \theta_1 + \theta_2 xy=θ1+θ2x

theta1theta_1theta1

theta2theta_2theta2

Where:

- yyy = predicted house price
- xxx = input feature
- θ1\theta_1θ1 = intercept
- θ2\theta_2θ2 = slope/weight

Working of the Practical

Step 1: Load Dataset

Boston Housing dataset is loaded using sklearn.

Step 2: Convert to DataFrame

Data is converted into pandas dataframe for easy handling.

Step 3: Split Dataset

Dataset divided into:

- Training data (80%)
- Testing data (20%)

Purpose:

- Train on one set
- Test on unseen data

Model Training

LinearRegression() model is used.

The model:

1. Learns patterns from training data
2. Finds best-fit line
3. Predicts house prices

Cost Function

Error is calculated using Mean Squared Error (MSE).

MSE=1n∑i=1n(yi−yi^)2MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y_i})^2MSE=n1∑i=1n(yi−yi^)2

Smaller MSE = Better predictions.

Gradient Descent

Gradient Descent updates weights to minimize error.

Process:

1. Start with random weights
2. Calculate error
3. Update weights
4. Repeat until minimum loss

Output

Scatter graph compares:

- Actual prices
- Predicted prices

Closer points indicate better accuracy.

Conclusion

The model successfully predicts Boston housing prices using regression techniques and neural network concepts.

Expected Viva/Oral Questions with Answers

1. What is Linear Regression?

Linear Regression is a supervised learning algorithm used to predict continuous values using a linear relationship between input and output.

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2. What is supervised learning?

In supervised learning, the model learns from labeled data containing both inputs and correct outputs.

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3. What is the role of Gradient Descent?

Gradient Descent minimizes the loss function by updating weights iteratively.

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4. What is MSE?

Mean Squared Error measures the average squared difference between predicted and actual values.

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5. Why is Boston Housing dataset used?

It is a standard regression dataset used for predicting house prices.

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6. Difference between classification and regression?

- Regression predicts continuous values.
- Classification predicts categories/classes.

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7. Why do we split train and test data?

To evaluate model performance on unseen data.

8. What is overfitting?

When the model memorizes training data and performs poorly on new data.

9. What are epochs?

One complete pass of training data through the model.

10. What is a neural network?

A neural network is a computational model inspired by the human brain that learns patterns from data.